
INSTITUTE OF COMPUTING – UNICAMP

MO903 – Scientific Writing

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Assignment 3 – Literature Review

Intent Mediation in SDN Networks

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1 Introduction

Software-Defined Networking (SDN) and Intent-Driven Networking (IDN) have been used to increase programmability and central control in computer networks (Herbaut et al., 2021; Alonso-Lupez et al., 2022). In SDN, the control plane is separated from the data plane and exposed through controller APIs, which allows the deployment of security and automation mechanisms that react to network events (Ahmad et al., 2023; Dungarani and Gujjar, 2024). In IDN, operators or applications can express *intents* that describe what the network should achieve, while the controller chooses how to implement them (El-Rif et al., 2023; Wang et al., 2025).

Current controllers such as ONOS provide intent frameworks, but still require manual configuration of several parameters, such as host identifiers, path constraints and policy details (Hyder and Fatima, 2021; El-Rif et al., 2023). This manual translation from operational goals to controller artefacts can lead to misconfigurations and slows down reactions in dynamic environments, for example in IoT deployments under DDoS attacks (Ahmad et al., 2023; Kim et al., 2024). The literature on SDN security shows that programmable control can support dynamic firewalls, intrusion detection and mitigation, but also that these mechanisms depend on human effort to define and maintain rules and policies (Abassi and Tabbane, 2023; Zhang et al., 2025).

At the same time, work on Large Language Models (LLMs) and natural-language interfaces for networking suggests that AI systems can help interpret requirements and generate configuration suggestions (Minhas et al., 2024; EROL and DEMİRCİ, 2025). Some studies explore LLMs and related models to support configuration, diagnosis or policy management in networks (Kou et al., 2025; Parajuli et al., 2025). Other studies focus on the robustness and security risks of LLMs, such as prompt injection and non-deterministic behaviour (Lin et al., 2025; Abouelseoud and El-Aasser, 2025). These results indicate both the potential and the risks of using LLMs as intermediaries between human operators and network controllers.

This work presents a systematic literature review (SLR) on security and autonomy in SDN/IDN environments, with a focus on LLM-mediated intent translation. The review follows a protocol inspired by PRISMA 2020, uses a PICO-style formulation to define the research questions and documents the search, screening and selection steps. From 68 records identified in IEEE Xplore and the ACM Digital Library, 54 unique studies were screened, 26 were analysed in full text and 24 were selected for synthesis. The selected works cover SDN-based security mechanisms, intent-driven orchestration frameworks and studies that explore AI or LLMs in network management and security (Herbaut et al., 2021; Ahmad et al., 2023; El-Rif et al., 2023; Alonso-Lupez et al., 2022; Minhas et al., 2024; Lin et al., 2025).

The SLR has four goals. The first goal is to describe how security and automation appear in SDN and IDN architectures. The second is to identify limitations in the translation of high-level intents into controller configurations. The third is to map how NLP and LLM-based approaches are used in this context. The fourth is to highlight open problems for a mediator architecture that integrates LLMs into SDN/IDN control planes in a controlled way.

2 Method

This section describes the protocol, the research questions, the search strategy and the selection process used in the SLR. The method follows a three-phase protocol and a PRISMA-style flow that can be seen in Figure 1.

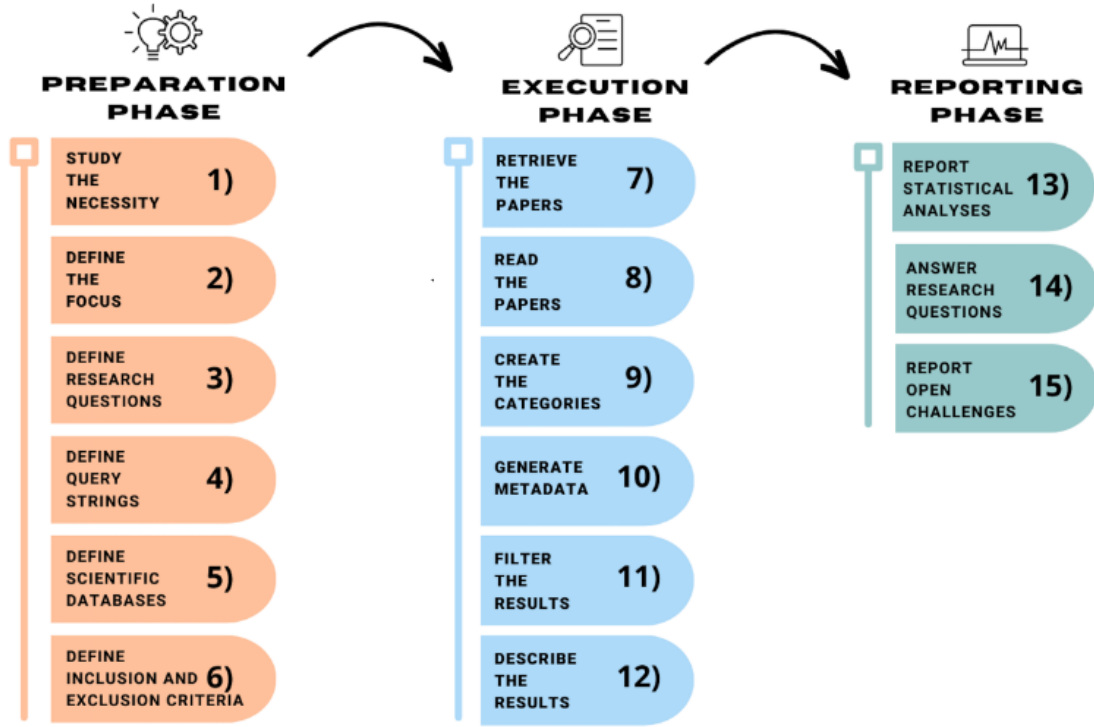


Figure 1: Overview of the three-phase SLR protocol used in this review, adapted from Regino et al. (2024)

2.1 Protocol and Research Questions

The protocol is based on PRISMA 2020 and was adapted to the scope and time constraints of the MO903 activity. The review uses a PICO structure. The population consists of SDN and IDN networks, with focus on IoT and edge scenarios (Herbaut et al., 2021; Alonso-Lupez et al., 2022). The intervention is the use of automation mechanisms that rely on intents, AI, NLP or LLMs to configure or adapt the network (El-Rif et al., 2023; Minhas et al., 2024). The comparison is manual configuration or non-intent-based automation. The outcomes are changes in security, autonomy and transparency of control decisions.

The main research question is:

How can large language models (LLMs) mediate the translation of high-level intents into SDN/IDN controller configurations to enhance security and autonomy in IoT networks?

From this question, the review derives five subquestions. The first subquestion asks how SDN-based approaches use automation or intent-based paradigms to improve security (Ahmad et al., 2023; Dungarani and Gujjar, 2024). The second asks how IDN models support autonomous and adaptive management (El-Rif et al., 2023; Wang et al., 2025). The third asks which limitations exist in the translation of high-level intentions into SDN controller configurations, especially in ONOS and related platforms (Hyder and Fatima, 2021; de Trizio et al., 2024). The fourth asks how NLP and LLM models can help bridge the gap between network intents and low-level configurations (Minhas et al., 2024; Lingga et al., 2025). The fifth asks which security implications arise from integrating LLMs into SDN/IDN control planes (Lin et al., 2025; Rivera et al., 2020).

2.2 Preparation Phase

The preparation phase started with a short exploratory reading of SDN security, IDN and AI-for-networking. This reading confirmed that SDN controllers already support dynamic security mechanisms, but that the mapping from operational goals to controller rules is still manual (Ahmad et al., 2023; Abassi and Tabbane, 2023). It also showed that IDN frameworks introduce higher-level abstractions, but still depend on predefined intent types and manual integration with applications (Hyder and Fatima, 2021; El-Rif et al., 2023). Work on LLMs in networking and security indicated that models can generate configuration suggestions, but also that they introduce new failure modes (Minhas et al., 2024; Lin et al., 2025).

Based on this preparation, the review scope was limited to studies that deal with SDN or IDN security, network automation with AI or ML, natural-language interfaces for network management or LLM-based mediation. Studies that focus only on hardware or optical aspects, without controller logic, were excluded.

2.3 Search Strategy and Databases

The search strings combined SDN, IDN, security and AI-related terms. A general expression used in the protocol was:

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("Software Defined Networking" OR SDN) AND ("Intent Based Networking"
OR "Intent Driven Network" OR IDN) AND (security OR "network protection"
OR "self-protective" OR "autonomous adaptation")
AND ("artificial intelligence" OR "machine learning"
OR "large language model" OR LLM OR "natural language")
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Variations of this expression emphasised LLMs and network automation, ONOS and intent frameworks, or IoT security with dynamic adaptation. For IEEE Xplore, the query used the “All Metadata” field. For the ACM Digital Library, a similar string was used with the default search fields.

The review used two databases: IEEE Xplore and the ACM Digital Library. A pilot search showed that these two sources include the main SDN, IDN and AI-for-networking venues, which was sufficient for this activity.

2.4 Inclusion and Exclusion Criteria

The inclusion criteria were: publication between 2020 and 2025, English language, peer-reviewed venue, focus on SDN or IDN or network automation, and presence of at least one element related to security, adaptation or intent handling (Herbaut et al., 2021; Ahmad et al., 2023). The exclusion criteria were: non-technical texts such as editorials or tutorials without method, studies that do not address security or intents, and work that focuses only on hardware or physical layer.

2.5 Quality Assessment

In addition to screening studies against the scope of the review, a light-weight quality assessment was applied to all records that reached the title–abstract and full-text stages. The assessment combined explicit inclusion and exclusion criteria with a check on conceptual and methodological clarity.

Table 1 summarises the inclusion and exclusion criteria used during screening and quality assessment. Studies were included if they satisfied all inclusion conditions and did not fall under any exclusion category. During full-text reading, these criteria were revisited to confirm that each paper effectively addressed SDN or IDN control, security or adaptation, and some form of automation or AI-mediated orchestration. Papers that were only loosely related to security or intent frameworks, or that focused exclusively on hardware without a clear network-control context, were excluded at this stage.

Table 1: Inclusion and exclusion criteria used for screening and quality assessment.

Criterion type	Description
Inclusion	(a) Studies published between 2020–2025; (b) Written in English; (c) Peer-reviewed journal or conference papers; (d) Focus on SDN, IDN, network automation, or AI-mediated orchestration; (e) Present security, adaptation, or intent-related evaluation.
Exclusion	(a) Non-technical articles (e.g., editorials, tutorials without methodology); (b) Papers unrelated to security or intent frameworks; (c) Works focusing exclusively on hardware implementations without network control context; (d) Redundant or low-citation grey literature.

Beyond these binary criteria, the full-text studies were also assessed qualitatively in terms of (i) alignment with the main research question, (ii) clarity of the proposed approach or model, and (iii) presence of experiments, simulations or concrete examples of network behaviour. This procedure ensured that the final set of 24 studies combined conceptual relevance with a minimum level of methodological transparency.

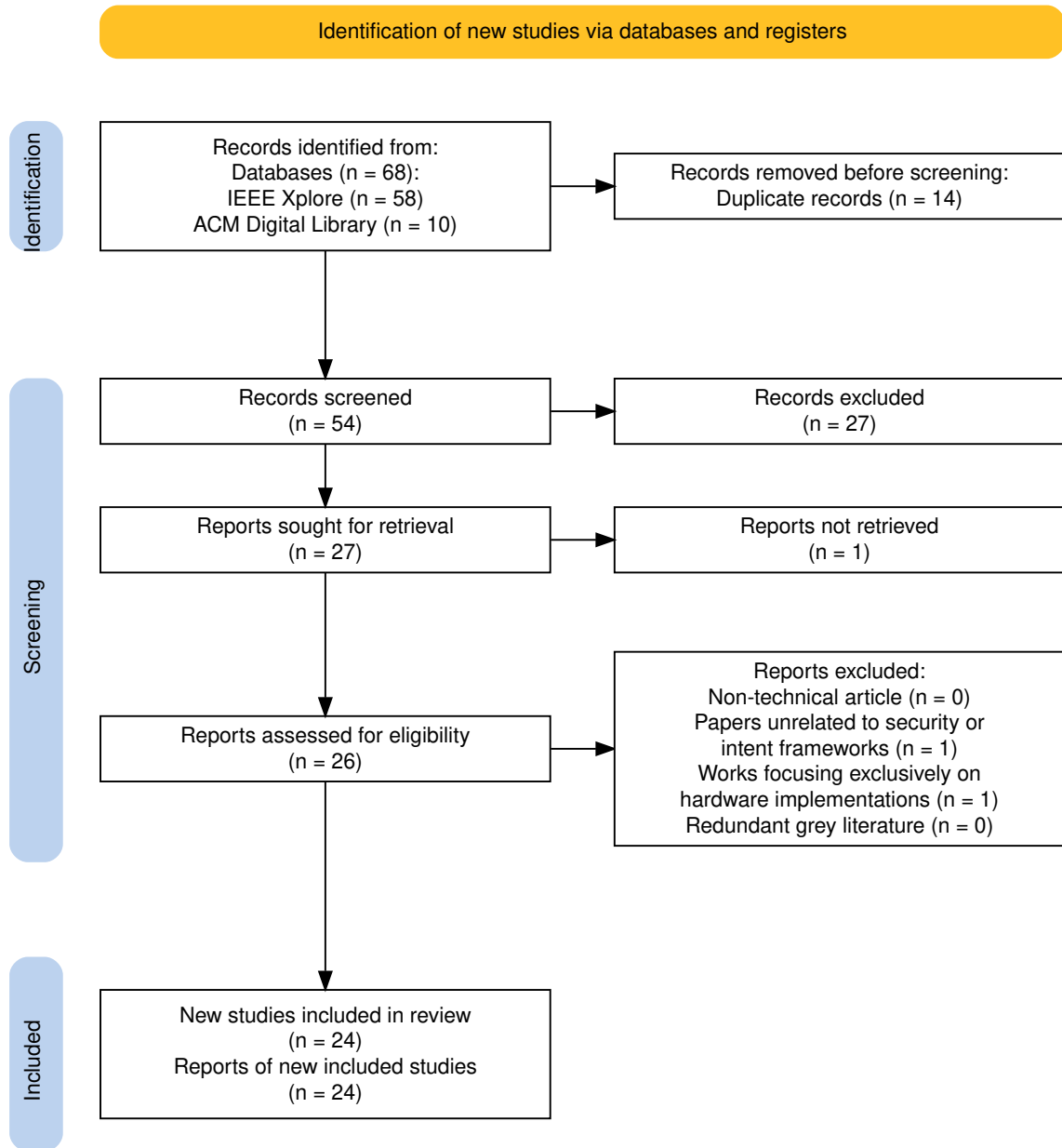


Figure 2: PRISMA 2020 flow diagram summarizing the identification, screening, eligibility and inclusion stages of the review, adapted from Haddaway et al. (2022).

2.6 Selection Process

The selection process followed four stages, as shown in Figure 2. In the identification stage, the search strings returned 68 records across the two databases. After importing the results into Zotero and applying deduplication, 54 unique records remained.

In the screening stage, titles and abstracts of the 54 records were checked against the criteria. Each record was marked according to four questions: whether it used SDN or IDN, whether it mentioned security, whether it presented a threat, attack or defence

mechanism, and whether it involved controller logic, intents or IoT. A simple score combined these marks and helped prioritise reading, but the decision to keep or remove a record was manual. At the end of this stage, 26 records remained.

In the eligibility stage, these 26 records were analysed in full text. The reading focused on conceptual relevance to SDN/IDN security and automation, clarity of contribution, and presence of experiments or simulations. Two studies were found to be only tangentially related to security or intents and were set aside, and 24 were kept.

In the inclusion stage, the 24 studies formed the main corpus for synthesis and discussion. They include SDN security mechanisms (Ahmad et al., 2023; Dungarani and Gujjar, 2024), IDN and intent-based frameworks (El-Rif et al., 2023; Alonso-Lupez et al., 2022), and work that explores AI or LLMs to support network management and security (Minhas et al., 2024; Lingga et al., 2025).

3 Results and Discussion

This section describes the main patterns found in the 24 selected studies and relates them to the research questions.

3.1 General Characterization

The studies were published between 2020 and 2025 and appear in SDN and networking venues, security conferences and AI or systems tracks. Several works use SDN controllers to deploy dynamic security mechanisms, such as firewalls, intrusion detection and DDoS mitigation (Ahmad et al., 2023; Dungarani and Gujjar, 2024). Others propose context-aware or attribute-based access control using SDN (Abassi and Tabbane, 2023; Zhang et al., 2025). A second group focuses on intents and higher-level policies in ONOS and similar frameworks (Hyder and Fatima, 2021; El-Rif et al., 2023). A third group explores AI and LLM-based support for configuration or analysis tasks, sometimes outside a strict SDN context but with clear links to network management (Minhas et al., 2024; Lingga et al., 2025).

3.2 Security in SDN with Automation and Intents (RQ1)

For RQ1, the SDN-based security studies show that the controller can act as a central point to detect and mitigate threats. Some works implement anomaly detection in the control plane and automatically install rules to block or reroute malicious flows (Ahmad et al., 2023; Dungarani and Gujjar, 2024). Others define context-aware policies that enforce restrictions based on user attributes or network conditions (Abassi and Tabbane, 2023; Zhang et al., 2025). These examples confirm that SDN enables dynamic and programmable defences.

In some cases, these policies are close to intents, even if they are not named as such. They describe desired states (for example, isolation of suspicious hosts or guaranteed paths for critical services) rather than individual flow rules. However, the mapping from these policies to the low-level configuration is still implemented manually in controller applications.

3.3 IDN Models and Autonomous Management (RQ2)

For RQ2, IDN and intent-based models define abstractions that represent connectivity, QoS and sometimes security constraints. Works on ONOS intents show how operators can specify host-to-host connectivity with constraints, while the controller computes paths and manages failures (Hyder and Fatima, 2021; El-Rif et al., 2023). Recent proposals extend these ideas to multi-layer or multi-domain settings and discuss higher-level intent hierarchies (Alonso-Lupez et al., 2022; Wang et al., 2025).

These models include monitoring and feedback loops. When an intent cannot be satisfied or is violated, the controller can adjust routes or allocate more resources. This behaviour supports more autonomous management. However, the vocabulary of intents is still fixed, and there is no general mechanism to turn informal security goals into concrete intent objects.

3.4 Limitations in Intent Translation (RQ3)

For RQ3, the main limitation is the absence of a systematic process that links human or application-level goals to controller artefacts. Even with an intent framework in ONOS, operators must understand the available intent types and configure them by hand (Hyder and Fatima, 2021; El-Rif et al., 2023). Security requirements are often expressed as additional fields or constraints, not as first-class properties, which makes their reuse and validation harder (Abassi and Tabbane, 2023; Alonso-Lupez et al., 2022).

Moreover, few works provide formal checks to verify that the resulting configuration is consistent with the original security goal. Some proposals analyse policy conflicts or generate explanations for intent behaviour, but these mechanisms are still specific to the frameworks where they were introduced (Alonso-Lupez et al., 2022; de Trizio et al., 2024).

3.5 NLP and LLMs as Mediators (RQ4)

For RQ4, studies that use NLP or LLMs show that natural-language descriptions can be mapped to configuration suggestions or diagnostic actions. Some works propose LLM-based assistants that help operators search documentation, generate commands or interpret logs (Minhas et al., 2024; Lingga et al., 2025). Others use ML models, including LLMs, to classify traffic and recommend security responses (Kou et al., 2025; Parajuli et al., 2025).

These results indicate that LLMs can act as mediators between informal goals and technical artefacts. However, the integration with SDN or IDN controllers is still loose. In most cases, the AI system outputs text or code that a human must validate and apply. There is still no end-to-end architecture that connects natural-language intents to SDN/IDN intent frameworks with explicit validation steps.

3.6 Security Implications of LLM Integration (RQ5)

For RQ5, work on LLM security highlights risks that are relevant if these models are placed close to the control plane. Studies describe prompt injection and other attacks that can subvert model outputs, as well as difficulties in guaranteeing reproducibility and traceability of responses (Lin et al., 2025; Abouelseoud and El-Aasser, 2025). There

are also concerns about the use of LLMs in security workflows and the need for human oversight (Rivera et al., 2020; Minhas et al., 2024).

These concerns suggest that LLMs should not directly install rules or intents in SDN/IDN controllers. Instead, they should generate candidate intents or policies that pass through additional checks, such as static analysis, templates or policy consistency tests. The literature reviewed here does not yet define such a mediator in detail, but it offers building blocks from SDN security, intent frameworks and LLM-based tooling.

4 Conclusion

This work presented a systematic literature review on security and autonomy in SDN/IDN environments, with focus on the use of LLMs as mediators of intent translation. The review followed a PRISMA-inspired protocol. It identified 68 records, screened 54 unique studies, analysed 26 in full text and selected 24 for synthesis. The studies covered SDN security mechanisms, IDN and intent-based frameworks and AI or LLM-based support for network management and security (Ahmad et al., 2023; El-Rif et al., 2023).

The review shows that SDN is a suitable base for dynamic security mechanisms and that IDN provides higher-level abstractions for management. It also shows that the link between real-world security objectives and controller artefacts is still manual and ad hoc (Hyder and Fatima, 2021; Alonso-Lupez et al., 2022). Work on LLMs and NLP in networking indicates that these models can help interpret and generate configurations, but also that they introduce new risks which must be mitigated (Minhas et al., 2024; Lin et al., 2025).

A central conclusion is that LLMs should be treated as assistants that help interpret and refine intents, not as autonomous controllers. A mediator architecture should separate the generation of candidate intents from their validation and enforcement. Future work can use the set of 24 studies identified here as a base to specify that mediator in more detail, combining SDN security patterns, intent models and safeguards for LLM use in security-critical environments.

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